

Zhiliang (Leon) Xiang

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Cardiff University, Senghennydd Rd, Cardiff CF24 4AG, UK

EDUCATION

- **Cardiff University** Jan. 2023 - Present
PhD Student Cardiff, UK
 - **Thesis:** Logic-based System for Collective Entity Resolution
 - **Supervisor:** Yazmín Ibáñez-García, Víctor Gutiérrez-Basulto
 - **Advised by:** Meghyn Bienvenu, Gianluca Cima
- **Cardiff University** Oct. 2020 - Oct. 2022
MSc Advanced Computer Science with High Distinction Cardiff, UK
 - **Dissertation:** Neuro-symbolic Approaches for Abstract and Relational Visual Reasoning Tasks
 - **Supervisor:** Víctor Gutiérrez-Basulto
- **Guangxi University of Science and Technology** Sept. 2011 - Sept. 2015
B.Sc. in Information and Computing Science Guangxi, China

INTERNSHIP/RESEARCH VISIT

- **Institute of Intelligent Information Processing, Shanxi University** July. 2026
Topic: Enhancing Logical Consistency in Attributed Question Answering for LLM Taiyuan, Shanxi, China
- **RATIO Research Group, LaBRI, Université de Bordeaux** Mar. 2026 - June 2026
Topic: Certain/Brave Solutions for Logic-based Entity Resolution Bordeaux, France
 - Supervisor: Meghyn Bienvenu
- **Inoue Laboratory, National Institute of Informatics** Sept. 2024 - Mar. 2025
Topic: Logic-based Rule Learning for Entity Resolution Tokyo, Japan
 - Supervisor: Katsumi Inoue
- **Centre for Artificial Intelligence, Robotics and Human-Machine Systems (IROHMS)** Sept. 2021 - Dec. 2022
Topic: Entity Resolution via Answer Set Programming Cardiff, UK
 - Supervisor: Víctor Gutiérrez-Basulto

GRANTS/FUNDING

- **European Laboratory for Learning and Intelligent Systems (ELLIS) PostDoc Program** Oct. 2026 - Oct. 2028
Competitive funding, 12 out of 412. Funded by: ELLIS Institute Finland
- **COMSC International Collaboration Scheme 2026** July 2026
£2,500. Funded by: COMSC Cardiff University
- **COMSC PhD Scholarship 2023** 2023 - 2026
*with Oversea student fee waiver (£22,700*4) + stipend. Funded by: COMSC Cardiff University*
- **EDBT-INTENDED School on Data and Knowledge Student Grant** 2022
Funded by: CNRS France & University of Bordeaux

PROJECTS

- **Intelligent Handling of Imperfect Data (ANR Chair in AI)** 2023-2026
with the LaBRI, Université de Bordeaux (Participant).
- **Embedding Methods to Complex Logical Query Answering on Knowledge Graph** 2021-2025
with Uni. of Edinburgh and Shanxi Uni. (Participant).
- **The OECD's Artificial Intelligence and the Future of Skills (AIFS)** 2021-2022
with Uni. of Leeds & VRRAIN, UPV (Participant).

SELECTED PUBLICATIONS

C=CONFERENCE, J=JOURNAL

- [C.1] **Zhiliang Xiang**, Meghyn Bienvenu , Gianluca Cima , Víctor Gutiérrez-Basulto, Yazmín Ibáñez-García (2025). **Advances in Logic-Based Entity Resolution: Enhancing ASPEN with Local Merges and Optimality Criteria**. In *Proceedings of the 22nd International Conference on Principles of Knowledge Representation and Reasoning, KR 2025*.
- [J.1] Meghyn Bienvenu, Gianluca Cima, Víctor Gutiérrez-Basulto, Yazmín Ibáñez-García, **Zhiliang Xiang** (2025). **Recent Advances in Logic-Based Entity Resolution**. *Database Principle Column, SIGMOD Record, Volume 54*.
- [C.2] **Zhiliang Xiang**, Meghyn Bienvenu , Gianluca Cima , Víctor Gutiérrez-Basulto, Yazmín Ibáñez-García. (2024). **ASPEN: ASP-Based System for Collective Entity Resolution**. In *Proceedings of the 21st International Conference on Principles of Knowledge Representation and Reasoning, KR 2024*.
- [C.4] Zhiwei Hu, Víctor Gutiérrez-Basulto, **Zhiliang Xiang**, Ru Li, Jeff Z. Pan (2022). **Transformer-based Entity Typing in Knowledge Graphs**. In *proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, EMNLP 2022*.
- [C.5] Zhiwei Hu, Víctor Gutiérrez-Basulto, **Zhiliang Xiang**, Xiaoli Li, Ru Li, Jeff Z. Pan: (2022). **Type-aware Embeddings for Multi-Hop Reasoning over Knowledge Graphs**. In *proceedings of the Thirty-First International Joint Conference on Artificial Intelligence, IJCAI 2022*.

SELECTED PRESENTATIONS

- **ASP-Based System for Collective Entity Resolution**. Invited Talk at Potsdam Uni. KRR Seminar. Invited by Torsten Schaub. Mar. 2026 [🌐]
- **Advances in Logic-Based Entity Resolution: Enhancing ASPEN with Local Merges and Optimality Criteria**. Conference Talk. KR 2025, Melbourne, Australia. Nov. 2025 [🌐]
- **Learning Rules for Entity Resolution: A Refinement Operator-based Approach**. Workshop Talk. 第6回杉山CRESTミーティング, Kyoto Uni., Japan. Mar. 2025 [🌐]
- **ASPEN: ASP-Based System for Collective Entity Resolution**. Conference Talk, KR 2024, Hanoi Nov. 2024 [🌐]
- **Artificial Intelligence Approaches Towards Structured Data Cleaning**. Guest Tutorial, Data Science, Escuela Superior Politécnica del Litoral. Oct. 2022
- **Type-aware Embeddings for Multi-Hop Reasoning over Knowledge Graphs**. Conference Talk, IJCAI 2022, Vienna July. 2022 [🌐]

TEACHING EXPERIENCE

- **Knowledge Representation (MSc)** Autumn 2023, 2025
Teaching Assistant Cardiff Uni.
 - **Contents:** Propositional/First-order/Description Logic, Nonmonotonic Reasoning, Belief Revision
- **Principle of Machine Learning (MSc)** Autumn 2023
Teaching Assistant Cardiff Uni.
 - **Contents:** PAC Learning Theory, Machine Learning Algorithms

SERVICES

- **KR2026 Learning and Explanation Track**. Reviewer, Mar. 2026
- **Joint Workshop on XLoKR & ExCoS (Co-located with FLOC2026)**, PC Member, 2026
- **Machine Learning Journal (MLJ)**. Reviewer 2024
- **ACM Transactions on Knowledge Discovery from Data**. Reviewer 2023